

VIDEO AQUISITION METHOD

1. Video Capture Device

Video footage was captured using a smartphone - Apple iPhone 13 pro equipped with a rear wide camera. This device was selected because it is widely available, easy to use, and provides high-resolution video suitable for road surface inspection.

Camera specifications:

- Resolution: 4K (3840×2160)
- Frame rate capability: 59.98 FPS (~60 FPS)

2. Recording Parameters and Their Significance

a) Resolution

The video was recorded at a resolution of 3840×2160 (4K).

This high resolution was chosen to clearly capture small cracks and potholes on the road surface, providing detailed information for analysis while balancing file size and processing needs.

b) Frame Rate

The frame rate used was 59.98 FPS.

This high frame rate ensures smooth motion in the video, which helps maintain continuity and avoids motion blur while the camera or vehicle moves.

c) Duration of Video

Each video clip was captured in different durations.

This duration covers damaged bits of the road segment, ensuring enough data is captured for analysis.

3. Environmental Conditions

The videos were recorded under the following conditions:

- Time of day: evening
- Lighting: natural daylight
- Weather: damp
- Presence of shadows: none

4. Camera Motion and Placement

The camera was handheld at a height of average from the road surface, with orientation downward facing.

This setup ensured that the road surface was clearly captured throughout the recording.

5. Site Selection

Videos were recorded on rural roads.

These locations were chosen because they had visible damage such as cracks, and traffic conditions were average during recording.

6. Recording Process and Sample Videos

- A total of 30 video clips were recorded.
- The recording procedure involved capturing videos of road surfaces for analysis purposes.

Out of these, a few clips were rejected due to camera shake and motion blur.

7. Video Analysis and Data set Preparation

- Recorded videos were saved in MP4 (HEVC / H.265) format.
- In future work, **OpenCV** will be used to automate frame extraction. This allows frames to be extracted at any interval automatically, reduces human effort and errors, ensures consistent data set organization, and prepares the data set efficiently for road damage analysis or machine learning applications.

8. Data Annotation Process

OpenCV or other automated methods can be explored in future work to make frame extraction faster and easier. Using such tools, frames can be extracted automatically at any chosen interval, organized into folders with unique names, and labeled efficiently. Automated methods reduce human effort, minimize errors, and make it possible to handle large datasets, which is important for training machine learning models for road damage detection.

9. Ethical and Safety Considerations

- No identifiable people or license plates were captured.
- All recordings were performed safely, without obstructing traffic or violating road safety rules.

GROUP 09

K. G.C.Pahasara	ICT/2023/040
D. M.M.M.Mahanama	ICT/2023/042
R.F.Fatheena	ICT/2023/079
M.G.J.N.Wijesinghe	ICT/2023/039